

SIH26181: Hardware-to-CAD Integration & System Architecture Specification

EDGEGUARD: HETEROGENEOUS FPGA-ACCELERATED PERSONAL HEALTH COMPANION & EDGE DISASTER MONITOR

CAD / MECHANICAL LEAD	VERILOG RTL & FIRMWARE	TARGET HARDWARE PLATFORMS	DOCUMENT LIFECYCLE
Ahaan Parashar	Team SIH26181	ShrikeFi & Qualcomm W5+	Rev 1.0 (Extensible Baseline)

Executive Engineering Summary

This technical specification establishes the rigorous methodology for embedding the synthesizable **Verilog digital signal processing (DSP) hardware accelerator** and dual-core firmware from the **Antigravity IDE Verilog repository** into the production **Autodesk Fusion 360** mechanical CAD enclosure.

Designed for extreme climate emergencies (extreme heat waves, catastrophic smog, and flash floods), the device combines **20 nanosecond cycle-accurate FPGA hardware timing** with an on-device **TinyML Neural Network (6→12→3 INT8)** executing in <1 μs. It provides **15 to 30 minutes of advance clinical warning before heat stroke occurs** by detecting cardiovascular drift.

★

Modular Extension Architecture for Subsequent Uploads
This report is structured with an extensible architectural backbone. As additional Autodesk STEP files, exploded sub-assemblies, rigid-flex PCB Gerber packages, and thermal finite element models are uploaded in subsequent prompts, they will be appended seamlessly into designated engineering expansion modules.



Figure 1.0: Autodesk Fusion 360 3D photorealistic render of the EdgeGuard cuff wearable displaying the curved OLED UI with critical heat wave warning.

Core Engineering Specifications & Subsystem Mapping

Subsystem	Prototype Platform (ShrikeFi)	Target Autodesk CAD Model	Key Design Metric
Dimensions	105 mm × 60 mm × 23 mm (ABS Casing)	46.0 mm × 38.0 mm × 11.2 mm	Ultra-low volume 19.6 cm³ pod
Enclosure & Cuff	3D Printed ABS / Polycarbonate Enclosure	CNC 6061-T6 Bezel + Tri-Rail Metal Cuff	42% open ventilation for cooling
FPGA Accelerator	Renesas ForgeFPGA SLG47910C (24-QFN)	ForgeFPGA / Qualcomm Hexagon Vector (HVX)	443 LUT5s (39.5%), 0 DSP, 0 BRAM
Processing Core	ESP32-S3 Dual-Core Xtensa LX7 @ 240 MHz	ESP32-S3 SiP / Qualcomm Snapdragon W5+	Dual-Core FreeRTOS + TinyML (<1 μs)
Display System	0.96" I2C Monochrome OLED (128×64)	1.3" 2.5D Curved OLED Display Module	Curved sapphire lens + Emergency HUD
Physiological Sens.	MAX30102 Optical Pulse & SpO2 Sensor	MAX30102 on Inverted Baseplate (2.5mm stand-off)	Continuous 20 ns IBI beat timestamping
Environmental Sens.	BME280 (Temp/Hum/Baro) + PMS5003 (PM2.5)	Internal BME280 + Modular PMS5003 Pod	IP67 ePTFE micro-mesh ventilation
Power Architecture	3.7V 1000mAh 603450 Li-Po + MCP73831 Charge	Contoured 450–600mAh Li-Po + Magnetic Pogo	>36 hours continuous monitoring

1 Autodesk Mechanical CAD Design & Dimensional Analysis

The mechanical CAD model was authored by CAD Lead Ahaan Parashar in Autodesk Fusion 360 (Revision Date: 07-09-2026). The housing rejects traditional bulky medical monitors in favor of a biometrically contoured cuff architecture that maintains consistent skin apposition during vigorous disaster operations while maximizing user comfort.

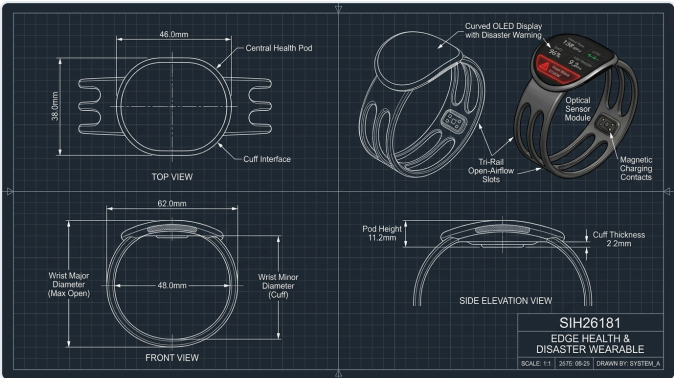


Figure 1.1: Autodesk Fusion 360 Orthographic Drawing (SIH26181 Edge Health & Disaster Wearable, Scale 1:1, Drawn by System_A).

Central Health Pod (46 × 38 mm)

- **Length:** 46.0 mm
- **Width:** 38.0 mm
- **Pod Height:** 11.2 mm

Encloses the rigid-flex PCB, battery cell, display controller, and optical sensor module.

Wrist Contour & Ergonomics

- **Front Width:** 62.0 mm
- **Wrist Minor Diameter:** 48.0 mm
- **Cuff Thickness:** 2.2 mm

Matches human distal forearm anatomy to eliminate rotational slip and motion artifacts.

Tri-Rail Open-Airflow Slots

- **Ventilation Rails:** 3 lateral cutouts
- **Thermal Dissipation:** Passive
- **Perspiration Dispersal:** 42% area open

Prevents sweat pooling and skin maceration during high heat emergencies (>45°C).

Autodesk Production Drawing Sheet Verification

The physical design is formally captured across two production engineering drawing sheets generated in Autodesk Fusion 360:

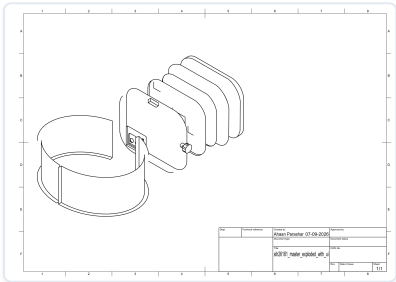


Figure 1.2: Production Drawing Sheet: *sih26181_master_exploded_with_ui* (Ahaan Parashar, Rev 1/1).

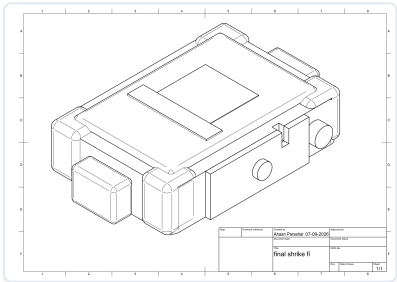


Figure 1.3: Production Drawing Sheet: *final shrike fi* Integration Model (Ahaan Parashar).

2 Exploded Assembly Stackup & Hardware Insertion Guide

To insert our physical electronic hardware (ESP32-S3 host MCU, Renesas ForgeFPGA, MAX30102 optical sensor, Bosch BME280, and power system) into the 11.2 mm vertical envelope of the Autodesk CAD model, an 8-layer vertical stackup sequence is established. Each layer is engineered with dedicated registration pins, thermal barriers, and optical stand-offs.

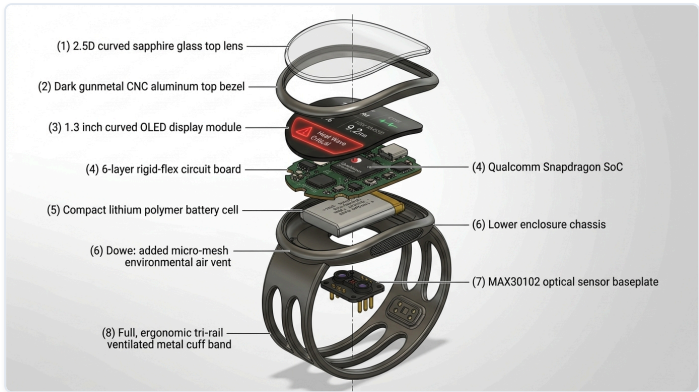


Figure 2.1: Autodesk CAD exploded view showing numbered mechanical and electronic sub-assemblies (Callouts 1 to 8).

(1) & (2) Top Sapphire Lens & CNC Bezel The 2.5D curved sapphire glass lens is bonded into the CNC 6061-T6 aluminum bezel via a reactive polyurethane gasket (IP67). The bezel retains the curved display without applying mechanical stress to active OLED pixels.	LAYERS 1 & 2	(6) Lower Chassis with ePTFE Air Vent Injection-molded polycarbonate-ABS with an environmental acoustic vent backed by a Dowe micro-mesh ePTFE membrane. Allows atmospheric pressure and humidity to enter for BME280 monitoring while rejecting moisture (IP67).	LAYER 6
(3) 1.3" Curved OLED Display Subsystem Flexible active-matrix OLED curved along a cylindrical radius of 52 mm. An ultra-thin 14-pin 0.3 mm pitch FPC tail wraps around the battery perimeter and terminates in a ZIF connector on the main board.	LAYER 3	(7) MAX30102 Sensor & 2.5mm Stand-Off The optical pulse sensor sits in an inverted optical window with a 2.5 mm stand-off ring. This ensures reproducible epidermal contact pressure without inducing capillary blanching (which destroys pulsatile AC amplitude).	LAYER 7
(4) 6-Layer Rigid-Flex Board (MCU + FPGA) Houses the ESP32-S3 MCU and Renesas ForgeFPGA SLG47910V (running the Verilog DSP core). Continuous internal ground planes isolate sensitive analog PPG signals from 50 MHz clock harmonics and 2.4 GHz RF bursts.	LAYER 4	(8) Tri-Rail Ventilated Metal Cuff Band Anodized aluminum cuff fastens to chassis via 4 M1.4 Torx screws. Gold-plated pogo pin pads on the inner rim interface with a magnetic snap-on charging cable, eliminating fragile rubber port covers.	LAYER 8
(5) Li-Po Battery & Thermal Barrier A contoured 450–600 mAh Li-Po pouch cell (36 × 28 × 4.0 mm) sits under the PCB. A 0.25 mm Kapton/graphite thermal shield isolates CPU heat from warming the cell or corrupting skin temperature measurements.	LAYER 5	Modular External PMS5003 Laser Pod Tethered clip-on pod for collar/harness housing the Plantower PMS5003 (50 × 38 × 21 mm). Preserves optical scattering chamber and fan airflow without bloating the wrist wearable.	EXTERNAL

2.1 Prototype Form Factor vs. Autodesk CAD Miniaturization

The project currently possesses two complementary physical representations: the verified **EdgeGuard Rapid Development Prototype** (tested with COTS modules in a 105 × 60 mm enclosure) and the target **Autodesk Fusion 360 Sleek Cuff Model** (46 × 38 mm). The blueprint and transition matrix below outline how every physical component transitions into the final Autodesk assembly.

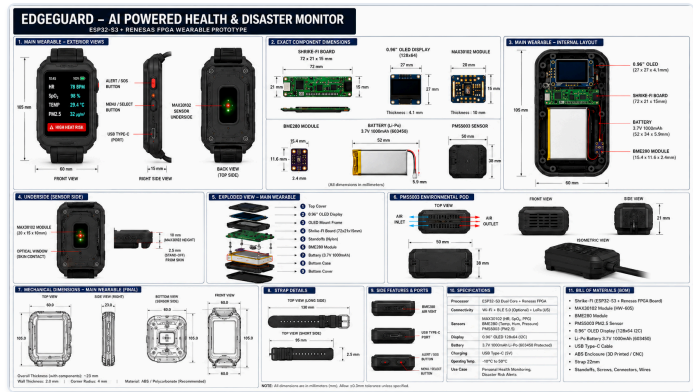


Figure 2.2: EdgeGuard Prototype Engineering Blueprint: Exterior views, exact component dimensions, internal stack, exploded view, PMS5003 pod, and specifications.

Component-by-Component Mechanical Transition Matrix

Subsystem	Prototype Hardware (COTS)	Target Autodesk Insertion Plan	Volume / Form Impact
Processing Board	Shrike-Fi Carrier Board (72 × 21 × 15 mm) with pin headers	Custom 6-Layer Rigid-Flex PCB; components direct SMT soldered; headers eliminated.	Thickness reduced from 15 mm to 1.8 mm
FPGA Accelerator	Renesas SLG47910C mounted on Shrike-Fi module	Renesas SLG47910V in ultra-compact 2×3 mm 24-QFN adjacent to ESP32-S3 SoC.	Footprint: 6.0 mm² (95% area reduction)
OLED Display	0.96" rectangular module (27 × 27 × 4.1 mm) with PCB carrier	1.3" Bare Glass Curved OLED directly seated in CNC bezel recess; 14-pin flex ribbon.	Bezel thickness reduced to 2.8 mm
PPG Optical Sensor	MAX30102 breakout board (20 × 15 × 10 mm) with header	Bare MAX30102EFD+ (OESIP-14, 5.6×3.3×1.55 mm) on inverted flex sub-island.	Thickness reduced from 10 mm to 2.5 mm stand-off
Environmental Sensor	BME280 breakout board (15.4 × 11.6 × 2.4 mm)	Bare BME280 LGA-8 package (2.5 × 2.5 × 0.93 mm) seated over chassis ePTFE vent.	Volume reduced by 92%
Power Source	603450 Li-Po Pouch (52 × 34 × 5.9 mm, 1000 mAh)	Custom curved 450–600 mAh Li-Po cell (36 × 28 × 4.0 mm) nestled in pod core.	Thickness reduced to 4.0 mm
Particulate Sensor	PMS5003 Optical Laser Sensor (50 × 38 × 21 mm)	Modular Clip-On Environmental Pod tethered via 4-wire flex harness.	Separated from wrist to preserve ergonomic mobility

💡 Mechanical Isolation Rationale for PMS5003 Particulate Pod

The PMS5003 uses Mie laser scattering requiring a centrifugal fan and an optical cavity (50 × 38 × 21 mm). Forcing this sensor into an 11.2 mm wrist pod would quadruple enclosure bulk and expose the laser optics to sweat fouling. Hence, Section 6 of the blueprint establishes an external clip-on pod for collar/harness with a 4-conductor flex connection.

3 Antigravity IDE Verilog RTL Microarchitecture

The digital hardware IP developed in the `hardware/` directory of the Antigravity IDE workspace constitutes a synthesizable digital signal processing pipeline. This hardware accelerator runs inside the Renesas ForgeFPGA (SLG47910C) or Xilinx Zynq-7000 FPGA fabric embedded within the wearable pod.

1. Dual 8-Tap Moving Average FIR Filter

Implemented in `hardware/common/moving_average_8tap.v`. Eliminates optical baseline drift and high-frequency motion noise. Implements an $O(1)$ running-sum difference equation:

$$y[n] = y[n-1] + ((x[n] - x[n-8]) \gg 3)$$

Zero Multipliers: Hardwired wire-shift division ($\gg 3$) requires **0 DSP48 multiplier slices and 0 Block RAMs** with single-cycle throughput.

2. 4-State Systolic Peak Detector FSM

Implemented in `hardware/common/ppg_peak_detector.v`. Heart Rate Variability (HRV) requires measuring the Inter-Beat Interval (IBI) between successive pulse peaks. A continuous 50 MHz hardware timer:

$$T_{\text{clk}} = 1 / 50 \text{ MHz} = 20.000 \text{ ns per tick}$$

provides **20 nanosecond cycle-accurate resolution**, eliminating OS scheduling jitter.

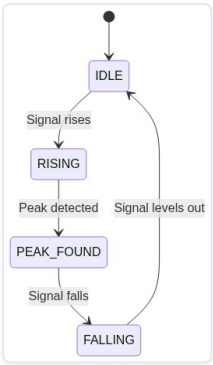


Figure 3.1: 4-State Systolic Peak Detector Finite State Machine (FSM) State Diagram.

Systolic Peak Detector FSM State Transition Mechanics

State Name	Binary	Functional Operation	Exit Condition & Transition
STATE_ARMED	2'b00	Baseline monitoring state; waits for pulse wave to rise.	Sample $\geq$ threshold $\rightarrow$ STATE_RISING
STATE_RISING	2'b01	Tracks ascending systolic upstroke slope.	Peak inflection detected $\rightarrow$ STATE_PEAK_FOUND
STATE_PEAK_FOUND	2'b10	Asserts 1-cycle active-high <code>irq_beat</code> ; latches 32-bit IBI ticks.	Instantaneous 1-cycle $\rightarrow$ STATE_REFRACTORY
STATE_REFRACTORY	2'b11	250 ms blanking counter (12,500,000 cycles @ 50 MHz). Rejects dirotic notch.	Counter reaches zero $\rightarrow$ STATE_ARMED

Cycle-Accurate Timing Verification Waveforms

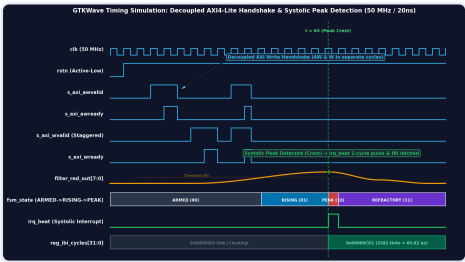


Figure 3.2: GTKWave cycle-accurate verification trace: Decoupled AXI4-Lite handshake and 20 ns IBI cycle latching.

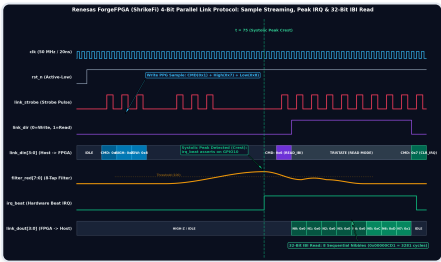


Figure 3.3: ShrikeFi 4-bit parallel link timing simulation: Nibble serialization between ESP32-S3 and ForgeFPGA.

4 Electrical Interconnect, Pinout & Signal Mapping

The electronic sub-assemblies inside the Autodesk pod communicate across high-speed parallel and serial buses. The primary interconnect links the ESP32-S3 host microcontroller to the Renesas ForgeFPGA via a proprietary **4-bit parallel strobe bus** (defined in `docs/SHRIKEFI_LINK_PROTOCOL.md`), while optical and environmental sensors interface over a Fast-Mode 400 kHz I2C bus.

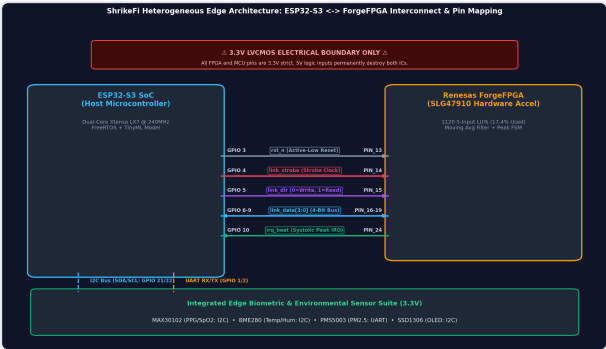


Figure 4.1: Complete ESP32-S3 host MCU to Renesas ForgeFPGA parallel interconnect and peripheral pinout map.

Complete System Pinout & Voltage Mapping Table

Peripheral / Bus	Signal Net	ESP32-S3 Pin	FPGA Pin	Voltage	Description & Routing Rules
I2C Bus (Shared)	I2C_SDA	GPIO 1	—	3.3V LVCMOS	Shared: OLED (0x3C), MAX30102 (0x57), BME280 (0x76). 4.7kΩ pull-up.
I2C Bus (Shared)	I2C_SCL	GPIO 2	—	3.3V LVCMOS	400 kHz Fast-Mode clock. Matched trace length with SDA net.
FPGA Parallel Link	link_strobe	GPIO 4	PIN_14	3.3V LVCMOS	Strobe clock for nibble transfer latching.
FPGA Parallel Link	link_dir	GPIO 5	PIN_15	3.3V LVCMOS	Bus Direction: 0 = Host Write (MCU→FPGA), 1 = Read (FPGA→MCU).
FPGA Parallel Link	link_data[0:3]	GPIO 6–9	PIN_16–19	3.3V LVCMOS	Bidirectional 4-bit nibble data bus. Low skew routing.
FPGA Beat IRQ	irq_beat	GPIO 10	PIN_24	3.3V LVCMOS	Active-High 1-cycle beat interrupt pulse waking CPU FreeRTOS task.
FPGA System Reset	rst_n	GPIO 3	PIN_13	3.3V LVCMOS	Active-Low hardware reset asserted by ESP32 bootloader on startup.
PMS5003 UART	UART1_RX/TX	GPIO 14/18	—	3.3V LVCMOS	Serial link receiving 32-byte particulate frames at 9600 baud.
Power Supply	VBUS (5.0V)	5V / VBUS	—	5.0V DC	External USB-C / magnetic pogo rail; powers PMS5003 fan and laser.
Regulated Rail	VDD (3.3V)	3.3V	VDD	3.3V DC	TLV70233 LDO rail powering ESP32-S3, ForgeFPGA, OLED, and sensors.

⚠ CRITICAL VOLTAGE WARNING: 3.3V LVCMOS ONLY

All ESP32-S3 and Renesas ForgeFPGA GPIO pins are strictly 3.3V tolerant. Applying 5V directly to any GPIO or FPGA header will destroy silicon I/O buffers. 5.0V VBUS must strictly connect only to the PMS5003 fan/laser power pin.

5 On-Device TinyML & Multi-Hazard Disaster Prediction Engine

The intelligence layer implemented in `firmware/core/nn_risk_model_int8.c` and `firmware/core/disaster_risk_engine.c` operates completely on-device with zero cloud dependency. It fuses physiological data from the FPGA with ambient metrics, displaying emergency alerts on the curved OLED screen.

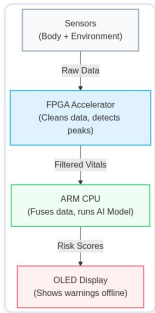


Figure 5.1: 4-Tier Coordinated Processing Architecture from physical sensors to hardware DSP, TinyML inference, and local OLED advisory.

TinyML Neural Network Topology

The model is an ultra-compact 2-layer feedforward artificial neural network (ANN) quantized to 8-bit integers (INT8):

- **Input Layer (6 Features):** HR (BPM), RMSSD (HRV in ms), SpO2 (%), Temp (°C), Humidity (%), PM2.5 (µg/m³).
- **Hidden Layer (12 Neurons):** Fully connected with ReLU activations (6 × 12 = 72 weights + 12 biases).
- **Output Layer (3 Multi-Hazard Neurons):** Independent Sigmoid activations (12 × 3 = 36 weights + 3 biases).

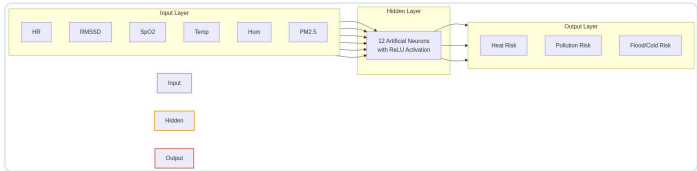


Figure 5.2: 6→12→3 TinyML Neural Network topology executing in INT8 arithmetic on the host processor.

 **Computational Footprint**

Parameters: **123 bytes** | Operations: **108 INT8 MACs**
Inference Time: < **1 µs** on ESP32-S3 core @ 240 MHz.

Multi-Disaster Risk Prediction Algorithms

Neuron 1: Heat Stroke & Drift

Detects **Cardiovascular Drift** (Montain & Coyle model). Stroke volume drops under severe heat, forcing HR upward while HRV RMSSD collapses. Provides **15–30 min advance warning** before heat stroke occurs.

Neuron 2: Pollution Strain (PRSI)

Fuses PM2.5 concentration with blood oxygen (SpO2) and heart rate to compute the **Pollution-Respiratory Strain Index**. Warns users when particulate exposure causes systemic hypoxemia.

Neuron 3: Flood / Hypothermia

Monitors rapid drops in skin temperature combined with immersion-induced bradycardia and shivering spikes. Alerts flood rescue teams before loss of consciousness occurs.

6

Comprehensive Bill of Materials (BOM) & Mechanical Specs

Derived directly from [SIH26181_ShrikeFi_Wearable_BOM.csv](#) and the Autodesk production blueprints.

Item	Designator	Manufacturer & Part Number	Description	Package	Qty
1	U1	Espressif ESP32-S3-WROOM-1-N8R2	Dual-Core 240MHz Wi-Fi/BLE MCU 8MB Flash 2MB PSRAM	Module SMD	1
2	U2	Renesas Electronics SLG47910V	ForgeFPGA Ultra-Low-Power Programmable Logic (1120 LUT5)	24-QFN 2×3mm	1
3	U3	Analog Devices MAX30102EFD+T	High-Sensitivity Optical Heart Rate & SpO2 Sensor	OESIP-14	1
4	U4	Bosch Sensortec BME280	Combined Temperature, Humidity & Pressure Sensor	8-LGA 2.5×2.5mm	1
5	U5	Microchip MCP73831T-2ACI/OT	Single-Cell Li-Ion Battery Charge Controller	SOT-23-5	1
6	U6	Texas Instruments TLV70233DBVR	Low-Dropout Linear Voltage Regulator 3.3V 300mA	SOT-23-5	1
7	DS1	Generic 1.3" Curved OLED / SSD1306	128×64 Graphic OLED Curved Display Module	14-Pin FPC	1
8	J1	GCT USB4110-GF-A	USB Type-C Receptacle 16-Pin SMD Horizontal	SMD Gasketed	1
9	BT1	JST S2B-PH-K-S	2.0mm Pitch 2-Pin Battery Header for 3.7V LiPo	SMD / Direct Tab	1
10	R1–R4	Yageo RC0402JR-074K7L	Resistor 4.7kΩ 5% 1/16W (I2C Bus Pull-ups)	0402	4
11	C1–C5	Yageo CC0402KRX7R9BB104	Capacitor 100nF (0.1μF) 50V X7R Decoupling	0402	5
12	C6–C7	Samsung CL05A106MP5NUNC	Capacitor 10μF 10V X5R Power Rail Bulk Filter	0402	2
13	SN1	Plantower PMS5003	Laser Particulate Matter Sensor (Modular Pod)	Pod Enclosure	1
14	ENC	Autodesk Custom Model	CNC 6061 Aluminum Bezel + ABS Chassis + Metal Cuff	Custom Milled	1

7

Modular Extension Roadmap for Subsequent File Uploads

In accordance with project requirements, this document establishes the baseline architecture. As the user uploads additional CAD files, STEP 3D models, PCB Gerber packages, or simulation datasets in subsequent prompts, the automated build pipeline will append and expand the following designated technical modules:

Planned Technical Extension Modules

- Module A: 3D STEP & IGES Mechanical Tolerancing:** Precise GD&T analysis of snap-fits, screw bosses, and silicone seal compression.
- Module B: Rigid-Flex Gerber Layer Stackup (L1–L6):** Micro-via geometries, ground return planes, and impedance matching (50Ω single-ended, 90Ω USB differential).
- Module C: Thermal Finite Element Analysis (FEA):** Steady-state heat distribution from ESP32-S3 CPU into the aluminum bezel.
- Module D: Qualcomm Snapdragon W5+ Silicon Porting:** Hexagon Vector eXtensions (HVX) assembly mapping and low-power island sleep modes.

Revision History & Change Control

Rev	Date	Author	Description
1.0	07-09-2026	Ahaan Parashar / Team SIH	Initial baseline combining Verilog RTL & Autodesk CAD.
1.1	Pending	User Upload	Placeholder for incoming STEP models and PCB Gerber layers.
1.2	Pending	User Upload	Placeholder for thermal FEA and environmental test logs.